

The Quiver associated to the category of G_n -equivariant perverse sheaves on $M_n(\mathbb{C})$

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1 Introduction

Many categories of perverse sheaves can be described as categories of representations of quivers with relations [GMV96; Vil94], although explicit descriptions are often difficult to obtain.

This article treats one of them. Let $X_n := M_n(\mathbb{C})$ be the variety of $n \times n$ complex matrices and let $G_n := GL_n(\mathbb{C}) \times GL_n(\mathbb{C})$ act on X_n by $(g, h) \cdot x = gxh^{-1}$. The orbits of this action are the sets X_n^k of matrices of corank k , for $0 \leq k \leq n$ which gives a stratification of the variety X_n . Braden and Grinberg [BG99] studied perverse sheaves with respect to this stratifications with coefficients in $\mathbb{C}$. Later, and in a different but parallel setting, Lőrincz and Walther [LW19] showed that the category of G_n -equivariant coherent $\mathcal{D}$ -modules on X_n is equivalent to the category of representations of the quiver $\widehat{AA}_n$:

$$0 \begin{array}{c} \xrightarrow{u_0} \\ \xleftarrow{v_0} \end{array} 1 \begin{array}{c} \xrightarrow{u_1} \\ \xleftarrow{v_1} \end{array} 2 \begin{array}{c} \xrightarrow{u_2} \\ \xleftarrow{v_2} \end{array} \cdots \begin{array}{c} \xrightarrow{u_{n-1}} \\ \xleftarrow{v_{n-1}} \end{array} n$$

with the relations

$$u_i v_i = 0 = v_i u_i, \quad 0 \leq i \leq n-1.$$

Let $\mathbb{k}$ be a field. Denote the category of G_n -equivariant perverse sheaves on X_n with coefficients in $\mathbb{k}$ by $\text{Perv}_{G_n}(X_n, \mathbb{k})$. It is a natural question to ask whether the quiver description exists for $\text{Perv}_{G_n}(X_n, \mathbb{k})$. Cipriani and Woolf [CW22] shows the quiver description exists as long as there are finitely many strata and the category of local systems on each of them has enough projectives, which is true in our case.

We show that there is a unique simple object associated to each stratum, denote it by IC_k for $0 \leq k \leq n$. Since the number of vertices are the number of simple objects, we get $n+1$ vertices of the quiver. The dimension of Ext^1 groups between the simple objects gives us all the arrows. The relations between the arrows are not easy to obtain and usually needs information about multiplicities of simples in projective covers. In our case, we will show how to get all the relations inductively. This gives the following main result:

Theorem 1.1: Let $\mathbb{k}$ be a field. For every $n \geq 1$ there is an equivalence of categories

$$\text{Perv}_{G_n}(X_n, \mathbb{k}) \cong \text{rep}(\widehat{AA}_n, \mathbb{k}).$$

The Theorem 1.1 lets one transport structural facts about $\text{rep}(\widehat{AA}_n)$, where everything is computable by hand, back to the geometric side.

1.1 Applications

The Fourier–Sato transform $\mathbb{T}$ is a t -exact autoequivalence of the category $\text{Perv}_{G_n}(X_n, \mathbb{k})$ that carries the constant sheaf IC_0 to the skyscraper IC_n . Its effect on the other simple objects are harder to compute in general. We will show how the uniseriality of composition factors in projective covers of IC_0 gives

$$\mathbb{T}(\text{IC}_k) \cong \text{IC}_{n-k}, \quad 0 \leq k \leq n.$$

Secondly, it's easier to construct the projective cover of simple objects from the quiver side. Thus, we can construct the projective cover of IC_k where each simple occurs exactly once. For $0 \leq k \leq n$, we will show that the perverse sheaf

$${}^p H^0(j_{0!}^k j_0^{k*} (\mathbb{T}({}^p H^0(j_{n-k!} \mathbb{k}[\dim X_n^{n-k}])))$$

is the projective cover of IC_k in $\text{Perv}_{G_n}(X_n, \mathbb{k})$. where $j_k : X_n^k \hookrightarrow X_n$ and $j_0^k : \bigcup_{i=0}^k X_n^i \hookrightarrow X_n$ are the inclusion maps.

2 Preliminaries

Let $X_n := M_n(\mathbb{C})$ and $G_n := \text{GL}_n(\mathbb{C}) \times \text{GL}_n(\mathbb{C})$ be the group acting on X_n by the following action

$$(g, h) \cdot x = gxh^{-1}$$

for all $(g, h) \in G_n$ and $x \in X_n$. The orbits of this action are completely classified by the corank of the matrix. This gives a stratification of X_n in terms of corank of matrices, i.e.,

$$X_n = \bigsqcup_{0 \leq k \leq n} X_n^k$$

where $X_n^k = \{x \in X_n \mid \text{corank}(x) = k\}$. Note that the closure of the stratum X_n^k is

$$\overline{X_n^k} = \bigsqcup_{l \geq k} X_n^l$$

and the dimension of X_n^k and its closure are $\overline{X_n^k}$ is given by

$$\dim X_n^k = n^2 - k^2 = \dim \overline{X_n^k} \tag{1}$$

for all $0 \leq k \leq n$.

Lemma 2.1: The stabilizer of any matrix $x \in X_n$ is connected.

Proof. The stabilizer of $x \in X_n$ is the subgroup:

$$\text{Stab}(x) = \{(g, h) \in G_n \mid gx = xh\}$$

Suppose the matrix x has corank r . Every matrix of corank r lies in the G_n -orbit of

$$x_r = \begin{pmatrix} I_{n-r} & 0 \\ 0 & 0 \end{pmatrix}$$

where I_{n-r} is the identity matrix of size $n-r$. Since stabilizers of points in one orbit are conjugate, it suffices to show $\text{Stab}(x_r)$ is connected.

Let $g, h \in \text{GL}_n(\mathbb{C})$ be partitioned into the same block structure

$$g = \begin{pmatrix} A & B \\ C & D \end{pmatrix}, \quad h = \begin{pmatrix} P & Q \\ R & S \end{pmatrix}$$

where A and P are $(n-r) \times (n-r)$ blocks. The stabilizer condition $gx_r = x_r h$ implies $A = P, C = 0$ and $Q = 0$.

Since g and h are invertible, their diagonal blocks A, D , and S must also be invertible. Thus, the stabilizer of x_r takes the form:

$$\text{Stab}(x_r) = \left\{ \left(\begin{pmatrix} A & B \\ 0 & D \end{pmatrix}, \begin{pmatrix} A & 0 \\ R & S \end{pmatrix} \right) \mid A \in \text{GL}_{n-r}(\mathbb{C}), D, S \in \text{GL}_r(\mathbb{C}), B, R^T \in \text{M}_{n-r,r}(\mathbb{C}) \right\}.$$

The Levi factor consisting of the block diagonal pairs determined by (A, D, S) forms a subgroup isomorphic to $\text{GL}_{n-r}(\mathbb{C}) \times \text{GL}_r(\mathbb{C}) \times \text{GL}_r(\mathbb{C})$. The unipotent radical consists of the pairs parameterized by B and R ; it is a normal subgroup isomorphic to the additive group of complex matrices $\text{M}_{n-r,r}(\mathbb{C}) \oplus \text{M}_{r,n-r}(\mathbb{C})$. Since the stabilizer is a semidirect product of two connected groups, the entire stabilizer group $\text{Stab}(x_r)$ is connected. $\square$

Let $\mathbb{k}$ be a field and $\text{D}_{\text{G}_n}^b(X_n, \mathbb{k})$ be the bounded derived category of G_n -equivariant complexes on X_n with coefficients in $\mathbb{k}$. Let $\text{Perv}_{\text{G}_n}(X_n, \mathbb{k}) \subset \text{D}_{\text{G}_n}^b(X_n, \mathbb{k})$ the full subcategory of G_n -equivariant perverse sheaves on X_n with coefficients in $\mathbb{k}$. Moreover, we will denote the constant sheaf on any space X by $\underline{\mathbb{k}}_X$.

Local systems are the building blocks of perverse sheaves. The following lemma describes the category of local systems on each stratum X_n^k .

Lemma 2.2: For $0 \leq k \leq n$, the category of local systems on X_n^k i.e., $\text{Loc}_{\text{G}_n}(X_n^k, \mathbb{k})$ is equivalent to the category of finitely generated $\mathbb{k}$ -modules.

Proof. Each stratum is a homogeneous G_n -space with connected stabilizers using the Lemma 2.1, so by [Ach21, Prop. 6.2.13], there are no non trivial G_n equivariant local system on X_n^k . $\square$

The above lemma shows that the category of local systems has enough projectives. This information is important for our next theorem.

Theorem 2.3: The category $\text{Perv}_{\text{G}_n}(X_n, \mathbb{k})$ has enough projectives.

Proof. Let

$$\begin{aligned} X_n^k &\hookrightarrow \overline{X_n^k}, \\ \overline{X_n^{k-1}} &\hookrightarrow \overline{X_n^k} \end{aligned}$$

be the open and the complementary closed embeddings. These induce a recollement of equivariant derived categories

$$\text{D}_{\text{G}_n}^b(X_n^k, \mathbb{k}) \longrightarrow \text{D}_{\text{G}_n}^b(\overline{X_n^k}, \mathbb{k}) \longrightarrow \text{D}_{\text{G}_n}^b(\overline{X_n^{k-1}}, \mathbb{k}),$$

as in [Bei+18, §1.4.3].

For $0 \leq k \leq n$, the heart of the category $D_{G_n}^b(X_n^k, \mathbb{k})$ is

$$\mathrm{Perv}_{G_n}(X_n^k, \mathbb{k}) \cong \mathrm{Loc}_{G_n}(X_n^k, \mathbb{k})[\dim X_n^k],$$

which is a length category with finitely many simple objects. Using the Lemma 2.2, it has enough projectives. Moreover, the heart of the category $D_{G_n}^b(\overline{X}_n^k, \mathbb{k})$ is $\mathrm{Perv}_{G_n}(\overline{X}_n^k, \mathbb{k})$, which is again a length category with finitely many simple objects for all $0 \leq k \leq n$.

We will use induction on k to show our result. For $k = 0$, the heart of $D_{G_n}^b(\overline{X}_n^0, \mathbb{k})$ is

$$\mathrm{Perv}_{G_n}(X_n^0, \mathbb{k}) \cong \mathrm{Loc}_{G_n}(X_n^0, \mathbb{k})[\dim X_n^0]$$

and has enough projectives.

Using induction of k and [CW22, Proposition 4.8], the heart of the glued t -structure on $D_{G_n}^b(\overline{X}_n^k, \mathbb{k})$, namely $\mathrm{Perv}_{G_n}(\overline{X}_n^k, \mathbb{k})$, has enough projectives. Taking $k = n$ gives the theorem. $\square$

When there are enough projective perverse sheaves and finitely many simple objects, the category of perverse sheaves can be described as modules over a finite-dimensional algebra, and therefore also as category of representations of a quiver with relations. The rest of the story is the quest to find this quiver.

3 Building the quiver

Let $\mathrm{rep}(\widehat{Q}, \mathbb{k})$ be the category of finite-dimensional $\mathbb{k}$ -representations of a quiver with relations $\widehat{Q} = (Q, I)$. The (isomorphism classes of) simple objects in $\mathrm{rep}(\widehat{Q})$ are in bijection with the vertices of Q . For each vertex x , the corresponding simple S^x is the representation with $(S^x)_x = \mathbb{k}$ and $(S^x)_y = 0$ for all other vertex $y \neq x$. Let P^x (resp. I^x) denote the projective cover (resp. injective envelope) of S^x , as constructed in [ASS06, §III.2]. In particular, for a vertex y , the dimension of $(P^x)_y$ (resp. $(I^x)_y$) is given by the number of paths from x to y (resp. from y to x), considered up to the relations in I .

We use this to recover the quiver from the category of perverse sheaves. The simple objects of the category will be in one-to-one correspondence with the vertices of the quiver. Moreover, the number of arrows from vertex x to vertex y will be calculated by finding the dimension of first extension group i.e., $\dim_{\mathbb{k}} \mathrm{Ext}^1(S^x, S^y)$ where S^x and S^y are the simple perverse sheaves corresponding to the vertex x and y respectively.

3.1 The vertices

The simple objects in $\mathrm{Perv}_{G_n}(X_n, \mathbb{k}) \subset D_{G_n}^b(X_n, \mathbb{k})$ are the complexes $\mathrm{IC}(X_n^k, L)$ where L runs over irreducible G_n -equivariant local systems of X_n^k . Using the Lemma 2.2, the only such irreducible local systems is the trivial local system.

Therefore,

$$\mathrm{IC}_k := \mathrm{IC}(X_n^k, \mathbb{K}_{X_n^k}[\dim X_n^k])$$

for $0 \leq k \leq n$, are all the simple objects up to isomorphism in $\mathrm{Perv}_{G_n}(X_n, \mathbb{K})$. This also tells us that there are $n + 1$ vertices in the quiver associated to $\mathrm{Perv}_{G_n}(X_n, \mathbb{K})$.

$$0 \quad 1 \quad 2 \quad \cdots \quad n$$

The vertex k in the above quiver corresponds to the simple perverse sheaf IC_k .

We will give an alternative description of the IC_k which will help us compute their stalks. Let

$$\widetilde{X}_n^k = \{(x, W) \in \overline{X}_n^k \times \mathrm{Gr}(k, n) \mid W \subset \ker(x)\}$$

where $\mathrm{Gr}(k, n)$ is the Grassmanian variety of k -dimensional spaces in $\mathbb{C}^n$. Let G_n act on $\widetilde{X}_n^k$ by

$$(g_1, g_2) \cdot (x, W) = (g_1 x g_2^{-1}, g_2 \cdot W)$$

for all $(g_1, g_2) \in G_n$. Under the usual $\mathrm{GL}_n(\mathbb{C})$ -action on $\mathrm{Gr}(k, n)$, the second projection $p_k : \widetilde{X}_n^k \rightarrow \mathrm{Gr}(k, n)$ is $\mathrm{GL}_n(\mathbb{C})$ -equivariant, and the fibers are linear and of the same dimension. This makes p_k a vector bundle over a smooth variety. Thus, $\widetilde{X}_n^k$ is a smooth variety for all k .

We stratify $\widetilde{X}_n^k$ by using the stratification on $\overline{X}_n^k$ i.e.,

$$\widetilde{X}_n^k = \bigsqcup_{l \geq k} \pi_k^{-1}(X_n^l)$$

where π_k is the first projection map.

Lemma 3.1: The first projection map $\pi_k : \widetilde{X}_n^k \rightarrow \overline{X}_n^k$ is proper and small with respect to X_n^k .

Proof. Consider the first projection map $\Pi_k : \overline{X}_n^k \times \mathrm{Gr}(k, n) \rightarrow \overline{X}_n^k$. Since $\mathrm{Gr}(k, n)$ is a projective variety, Π_k is proper. Moreover, $\widetilde{X}_n^k$ is a closed subvariety of $\overline{X}_n^k \times \mathrm{Gr}(k, n)$ which implies the restriction $\pi_k = \Pi_k|_{\widetilde{X}_n^k}$ is proper.

The biggest stratum of $\widetilde{X}_n^k$ is $\pi_k^{-1}(X_n^k) = X_n^k \times \mathrm{Gr}(k, k) \cong X_n^k$. Using 1, we get

$$\dim \widetilde{X}_n^k = n^2 - k^2.$$

For $l \geq k$ and $x_n^l \in X_n^l$, the fiber over x_n^l i.e., $\pi_k^{-1}(x_n^l)$ is isomorphic to $\mathrm{Gr}(k, l)$ which gives

$$\dim \pi_k^{-1}(x_n^l) = \dim \mathrm{Gr}(k, l) = k(l - k).$$

Note that X_n^k is an open dense subset in $\overline{X}_n^k$ and for $x_n^l \in X_n^l$ where $l > k$,

$$\dim \pi_k^{-1}(x_n^l) = k(l - k) < \frac{1}{2}(l^2 - k^2) = \frac{1}{2}(\dim \widetilde{X}_n^k - \dim X_n^l).$$

Thus, the map $\pi_k : \widetilde{X}_n^k \rightarrow \overline{X}_n^k$ is small with respect to X_n^k . □

Corollary 3.2: For $0 \leq k \leq n$,

$$\mathrm{IC}_k \cong \pi_{k*} \mathbb{K}_{\widetilde{X_n^k}}[\dim X_n^k]. \quad (2)$$

Proof. Since the maps π_k are small with respect to X_n^k and

$$\pi_k^{-1}(X_n^k) = X_n^k \times \mathrm{Gr}(k, k) \cong X_n^k,$$

the result follows from [Ach21, Prop. 3.8.10]. $\square$

It becomes easier to calculate stalks with the above description of IC objects. The following theorem gives a local description of our simple objects. Let ${}^p\mathrm{H}^i$ and ${}^p\tau^{\leq i}$ be the functors for the perverse cohomology and truncation respectively.

Theorem 3.3: For $n \geq l \geq k \geq 0$ and $t > -(l - k)^2$,

$$\begin{aligned} {}^p\mathrm{H}^t(\mathrm{IC}_k|_{X_n^l}, \mathbb{K}) &= 0 \\ {}^p\mathrm{H}^t(\mathrm{IC}_k|_{\overline{X_n^l}}, \mathbb{K}) &= 0. \end{aligned}$$

Proof. Using the isomorphism 2, for $x_n^l \in X_n^l$,

$$\mathrm{H}^t((\pi_{k*} \mathbb{K}_{\widetilde{X_n^k}}[\dim X_n^k])_{x_n^l}, \mathbb{K}) = \mathrm{H}_{\mathrm{singular}}^{t+\dim X_n^k}(\pi_k^{-1}(X_n^l), \mathbb{K}) = \mathrm{H}_{\mathrm{singular}}^{t+\dim X_n^k}(\mathrm{Gr}(k, l), \mathbb{K})$$

for all $k \leq l \leq n$. The top cohomology of $\mathrm{Gr}(k, l)$ lies in degree $2k(l - k)$ which implies

$$\mathrm{H}^t(\pi_{k*} \mathbb{K}_{\widetilde{X_n^k}}[\dim X_n^k]_{x_n^l}, \mathbb{K}) = 0$$

for $t > 2k(l - k) - \dim X_n^k = -\dim X_n^l - (l - k)^2$. This implies

$${}^p\mathrm{H}^t(\pi_{k*} \mathbb{K}_{\widetilde{X_n^k}}[\dim X_n^k]_{x_n^l}, \mathbb{K}) = \mathrm{H}^{t-\dim X_n^l}(\pi_{k*} \mathbb{K}_{\widetilde{X_n^k}}[\dim X_n^k]_{x_n^l}, \mathbb{K}) = 0$$

for $t > -(l - k)^2$.

Since $\overline{X_n^l} = \sqcup_{i \geq l} X_n^i$, and for $i \geq l$,

$${}^p\mathrm{H}^t(\pi_{k*} \mathbb{K}_{\widetilde{X_n^k}}[\dim X_n^k]_{X_n^i}, \mathbb{K}) = 0$$

for all $t > -(l - k)^2 \geq -(i - k)^2$, we get ${}^p\mathrm{H}^t(\pi_{k*} \mathbb{K}_{\widetilde{X_n^k}}[\dim X_n^k]_{\overline{X_n^l}}, \mathbb{K}) = 0$ for $t > -(l - k)^2$ from [Ach21, Lemma 3.1.6]. $\square$

In particular, for the inclusion map. $i_{l,k} : \overline{X_n^l} \hookrightarrow \overline{X_n^k}$, we have

$$i_{l,k}^* \mathrm{IC}_k \in {}^p\mathrm{D}_{\mathrm{G}_n}^b(\overline{X_n^l}, \mathbb{K})^{\leq -(l-k)^2}. \quad (3)$$

Since each IC_k is Verdier self dual, we get

$$i_{l,k}^! \mathrm{IC}_k \cong \mathbb{D}(i_{l,k}^* \pi_{k*} \mathbb{K}_{\widetilde{X_n^k}}[\dim X_n^k]) \in {}^p\mathrm{D}_{\mathrm{G}_n}^b(\overline{X_n^l}, \mathbb{K})^{\geq (l-k)^2}. \quad (4)$$

3.2 The arrows

In this section, we use the results above to compute $\text{Ext}^1(\text{IC}_k, \text{IC}_l)$ for $0 \leq k, l \leq n$.

Lemma 3.4: The first G_n -equivariant cohomology of X_n vanishes, i.e.,

$$H_{G_n}^1(X_n, \mathbb{k}) = \text{Hom}(\mathbb{k}_{X_n}, \mathbb{k}_{X_n}[1]) = 0.$$

Proof. Observe that X_n retracts G_n -equivariantly onto the zero matrix via the homotopy $(t, x) \mapsto tx$. Thus their equivariant cohomologies are isomorphic i.e.,

$$H_{G_n}^1(X_n, \mathbb{k}) \cong H_{G_n}^1(\{0_n\}, \mathbb{k}) \cong H^1(\text{BG}_n, \mathbb{k})$$

where BG_n is the classifying space of G_n . Since $G_n = \text{GL}_n(\mathbb{C}) \times \text{GL}_n(\mathbb{C})$, we get $\text{BG}_n \simeq \text{BGL}_n(\mathbb{C}) \times \text{BGL}_n(\mathbb{C})$. From [MS74, Theorem 14.5], the cohomology ring of $\text{BGL}_n(\mathbb{C})$ is a polynomial ring generated by the universal Chern classes

$$H^*(\text{BGL}_n(\mathbb{C}), \mathbb{k}) \cong \mathbb{k}[c_1, c_2, \dots, c_n],$$

where the topological degree of each Chern class is even (i.e., $\deg(c_i) = 2i$).

By the Künneth formula, $H^*(\text{BG}_n, \mathbb{k})$ is the tensor product of two such polynomial rings. Since all generators lie in even degrees, the cohomology groups in all odd degrees must vanish. Therefore, $H_{G_n}^1(X_n, \mathbb{k}) = 0$. $\square$

In a similar fashion, the same holds for each orbit X_n^k .

Lemma 3.5: The first G_n equivariant cohomology of X_n^k vanishes i.e.,

$$H_{G_n}^1(X_n^k, \mathbb{k}) = \text{Hom}(\mathbb{k}_{X_n^k}, \mathbb{k}_{X_n^k}[1]) = 0.$$

Proof. Using the orbit-stabilizer theorem, we have

$$X_n^k \cong G_n / \text{Stab}(x_k).$$

Moreover,

$$H_{G_n}^1(X_n^k, \mathbb{k}) \cong H_{G_n}^1(G_n / \text{Stab}(x_k), \mathbb{k}) \cong H^1(\text{BStab}(x_k), \mathbb{k})$$

where $\text{BStab}(x_k)$ is the classifying space of $\text{Stab}(x_k)$.

From 2, the unipotent radical of $\text{Stab}(x_k)$ i.e., $M_{n-k,k}(\mathbb{C}) \oplus M_{k,n-k}(\mathbb{C})$ is contractible. So, $\text{Stab}(x_k)$ retracts to its Levi subgroup i.e., $\text{GL}_{n-k} \times \text{GL}_k \times \text{GL}_k$. Thus,

$$H^*(\text{BStab}(x_k), \mathbb{k}) \cong H^*(\text{BGL}_{n-k}(\mathbb{C}), \mathbb{k}) \otimes H^*(\text{BGL}_k(\mathbb{C}), \mathbb{k}) \otimes H^*(\text{BGL}_k(\mathbb{C}), \mathbb{k})$$

by the Künneth formula. From [MS74, Theorem 14.5], the cohomology ring of $\text{BGL}_n(\mathbb{C})$ is a polynomial ring generated by the universal Chern classes

$$H^*(\text{BGL}_n(\mathbb{C}), \mathbb{k}) \cong \mathbb{k}[c_1, c_2, \dots, c_n],$$

where the topological degree of each Chern class is even (i.e., $\deg(c_i) = 2i$). Since all generators lie in even degrees, the cohomology groups in all odd degrees must vanish. Therefore, $H_{G_n}^1(X_n^k, \mathbb{k}) = 0$. $\square$

The following theorem tells us that there are no loops in our associated quiver.

Theorem 3.6: For $0 \leq k \leq n$,

$$\text{Ext}^1(\text{IC}_k, \text{IC}_k) = 0$$

Proof. Consider the distinguished triangle,

$$j_{k!} \mathbb{K}_{X_n^k}[\dim X_n^k] \rightarrow \text{IC}_k \rightarrow i_{k!} i_k^* \text{IC}_k \rightarrow,$$

where $j_k : X_n^k \hookrightarrow \overline{X_n^k}$ is the open embedding and i_k the complementary closed embedding. Applying the functor $\text{Hom}(-, \text{IC}_k[1])$ and adjoint relation, we get the following long exact sequence

$$\text{Hom}(\mathbb{K}_{X_n^k}[\dim X_n^k], \mathbb{K}_{X_n^k}[\dim X_n^k + 1]) \leftarrow \text{Ext}^1(\text{IC}_k, \text{IC}_k) \leftarrow \text{Hom}(i_k^* \text{IC}_k, i_k^! \text{IC}_k[1]) \leftarrow .$$

Using the Lemma 3.5, we have

$$\text{Hom}(\mathbb{K}_{X_n^k}[\dim X_n^k], \mathbb{K}_{X_n^k}[\dim X_n^k + 1]) = 0.$$

From 3 and 4, we can see that $i_k^* \text{IC}_k \in {}^p D_{G_n}^b(\overline{X_n^{k+1}}, \mathbb{K})^{\leq -1}$ and $i_k^! \text{IC}_k[1] \in {}^p D_{G_n}^b(\overline{X_n^{k+1}}, \mathbb{K})^{\geq 0}$ which implies

$$\text{Hom}(i_k^* \text{IC}_k, i_k^! \text{IC}_k[1]) = 0.$$

Thus,

$$\text{Ext}^1(\text{IC}_k, \text{IC}_k) = 0$$

for all $0 \leq k \leq n$. □

The following theorem tells us that there are no arrows between non consecutive vertices.

Theorem 3.7: For $0 \leq k, l \leq n$, and $l - k \geq 2$,

$$\text{Ext}^1(\text{IC}_k, \text{IC}_l) = 0$$

$$\text{Ext}^1(\text{IC}_l, \text{IC}_k) = 0$$

Proof. Let $i_{l,k} : \overline{X_n^l} \hookrightarrow \overline{X_n^k}$. Since IC_l is supported on $\overline{X_n^l} \subset \overline{X_n^k}$,

$$\begin{aligned} \text{Ext}^1(\text{IC}_k, \text{IC}_l) &= \text{Hom}(\text{IC}_k, i_{l,k*} \text{IC}_l[1]) \\ &= \text{Hom}(i_{l,k}^* \text{IC}_k, \text{IC}_l[1]) \\ &= 0 \end{aligned} \tag{by3}$$

and

$$\begin{aligned} \text{Ext}^1(\text{IC}_l, \text{IC}_k) &= \text{Hom}(i_{l,k*} \text{IC}_l, \text{IC}_k[1]) \\ &= \text{Hom}(i_{l,k}^! \text{IC}_l, \text{IC}_k[1]) \\ &= \text{Hom}(\text{IC}_l, i_{l,k}^! \text{IC}_k[1]) \\ &= 0. \end{aligned} \tag{by4}$$

Hence, for $0 \leq k, l \leq n$, and $l - k \geq 2$,

$$\text{Ext}^1(\text{IC}_k, \text{IC}_l) = 0$$

$$\text{Ext}^1(\text{IC}_l, \text{IC}_k) = 0.$$

□

Let

$$j_k : X_n^k \hookrightarrow X_n$$

be the inclusion map of the stratum X_n^k in X_n . We will construct the projective cover and injective envelope of IC_0 and IC_n .

Lemma 3.8: Let $0 \leq k \leq n-1$, $U_k := X_n^k \cup X_n^{k+1}$ and $f_k : U_k \rightarrow \overline{X_n^k}$ be the open embedding. Then,

$$\begin{aligned} {}^p H^0(f_{k*}(\mathrm{IC}_k|_{U_k})) &= \mathrm{IC}_k \\ {}^p H^1(f_{k*}(\mathrm{IC}_k|_{U_k})) &= 0 \end{aligned}$$

Proof. The map $i_{k+2,k} : \overline{X_n^{k+2}} \hookrightarrow \overline{X_n^k}$ is the complementary closed embedding to f_k . We have a distinguished triangle

$$i_{k+2,k*} i_{k+2,k}^! \mathrm{IC}_k \rightarrow \mathrm{IC}_k \rightarrow f_{k*}(\mathrm{IC}_k|_{U_k}) \rightarrow .$$

Since IC_k is perverse, only the zeroth perverse cohomology is nonzero and

$${}^p H^0(\mathrm{IC}_k) = \mathrm{IC}_k .$$

Using 4 and $i_{k+2,k*}$ is t -exact, we get

$${}^p H^t(i_{k+2,k*} i_{k+2,k}^! \mathrm{IC}_k) = 0$$

for $t \leq 3$. Taking the long exact sequence of perverse cohomology of the above distinguished triangle, we get the desired isomorphisms. $\square$

Theorem 3.9: Let $0 \leq k \leq n-1$, $h_k : X_n^{k+1} \hookrightarrow U_k$ be the closed embedding and $j_k : X_n^k \hookrightarrow \overline{X_n^k}$ be the open embedding.

$$\begin{aligned} {}^p H^0(j_{k+1*} h_k^!(\mathrm{IC}_k|_{U_k})) &= 0 \\ {}^p H^1(j_{k+1*} h_k^!(\mathrm{IC}_k|_{U_k})) &= {}^p H^0(j_{k+1*} \mathbb{K}_{X_n^{k+1}}[\dim X_n^{k+1}]) \end{aligned}$$

Proof. From 4, we have $h_k^!(\mathrm{IC}_k|_{U_k}) \in {}^p D^{\geq 1}$ and j_{k+1*} is left t -exact, which implies $j_{k+1*} h_k^!(\mathrm{IC}_k|_{U_k}) \in {}^p D^{\geq 1}$. Thus, ${}^p H^0(j_{k+1*} h_k^!(\mathrm{IC}_k|_{U_k})) = 0$.

Using the proper base change theorem, we can write

$$\mathrm{IC}_k|_{U_k} \cong \pi_{k*} \mathbb{K}_{\widetilde{X_n^k}}[\dim X_n^k]|_{U_k} \cong \pi'_{k*} \mathbb{K}_{\pi_k^{-1}(U_k)}[\dim X_n^k]$$

where $\pi'_k : \pi_k^{-1}(U_k) \xrightarrow{\pi_k} U_k$.

The term $h_k^! \pi_{k*} \mathbb{K}_{\widetilde{X_n^k}}[\dim X_n^k]|_{U_k} \cong h_k^! \pi'_{k*} \mathbb{K}_{\pi_k^{-1}(U_k)}[\dim X_n^k]$ can be further simplified using the following Cartesian square

$$\begin{array}{ccc} \pi_k^{-1}(X_n^{k+1}) & \xrightarrow{h'_k} & \pi_k^{-1}(U_k) \\ \pi''_k \downarrow & & \downarrow \pi'_k \\ X_n^{k+1} & \xrightarrow{h_k} & U_k \end{array}$$

to get

$$h_k^! \pi_{k*}' \mathbb{L}_{\pi_k^{-1}(U_k)}[\dim X_n^k] \cong \pi_{k*}'' h_k^! \mathbb{L}_{\pi_k^{-1}(U_k)}[\dim X_n^k].$$

Since $\pi_k^{-1}(U_k)$ is smooth and $\dim(\pi_k^{-1}(U_k)) = \dim X_n^k$, we can write

$$\begin{aligned} h_k^! \mathbb{L}_{\pi_k^{-1}(U_k)}[\dim X_n^k] &\cong h_k^! w_{\pi_k^{-1}(U_k)}[-\dim X_n^k] \\ &\cong w_{\pi_k^{-1}(X_n^{k+1})}[-\dim X_n^k] \\ &\cong \mathbb{L}_{\pi_k^{-1}(X_n^{k+1})}[2\dim(\pi_k^{-1}(X_n^{k+1})) - \dim X_n^k] \\ &\cong \mathbb{L}_{\pi_k^{-1}(X_n^{k+1})}[2(\dim X_n^{k+1} + k) - \dim X_n^k] \\ &\cong \mathbb{L}_{\pi_k^{-1}(X_n^{k+1})}[\dim X_n^{k+1} + 2k + \dim X_n^{k+1} - \dim X_n^k] \\ &\cong \mathbb{L}_{\pi_k^{-1}(X_n^{k+1})}[\dim X_n^{k+1} - 1] \end{aligned}$$

Since the fibers of π_k'' are isomorphic to $\mathbb{P}^k$ and j_{k+1*} is left t -exact, we have the following

$$\begin{aligned} {}^p H^1(j_{k+1*} h_k^!(IC_k|_{U_k})) &\cong {}^p H^1(j_{k+1*} \pi_{k*}'' h_k^! \mathbb{L}_{\pi_k^{-1}(U_k)}[\dim X_n^k]) \\ &\cong {}^p H^1(j_{k+1*} \pi_{k*}'' \mathbb{L}_{\pi_k^{-1}(X_n^{k+1})}[\dim X_n^{k+1} - 1]) \\ &\cong {}^p H^0(j_{k+1*} \pi_{k*}'' \mathbb{L}_{\pi_k^{-1}(X_n^{k+1})}[\dim X_n^{k+1}]) \\ &\cong {}^p \tau^{\leq 0}(j_{k+1*} \pi_{k*}'' \mathbb{L}_{\pi_k^{-1}(X_n^{k+1})}[\dim X_n^{k+1}]) \\ &\cong {}^p \tau^{\leq 0}(j_{k+1*} {}^p H^0(\pi_{k*}'' \mathbb{L}_{\pi_k^{-1}(X_n^{k+1})}[\dim X_n^{k+1}])) \\ &\cong {}^p \tau^{\leq 0}(j_{k+1*} \mathbb{L}_{X_n^{k+1}}[\dim X_n^{k+1}]) \\ &\cong {}^p \tau^{\leq 0}(j_{k+1*} {}^p H^0(\mathbb{L}_{X_n^{k+1}}[\dim X_n^{k+1}])) \\ &\cong {}^p \tau^{\leq 0} j_{k+1*}(\mathbb{L}_{X_n^{k+1}}[\dim X_n^{k+1}]) \\ &\cong {}^p H^0(j_{k+1*} \mathbb{L}_{X_n^{k+1}}[\dim X_n^{k+1}]) \end{aligned}$$

□

The following theorem will describe composition series of some important perverse objects.

Theorem 3.10: Let $0 \leq k \leq n-1$ and $j_k : X_n^k \hookrightarrow \overline{X_n^k}$ be the inclusion map. Then, we get the following exact sequences

$$0 \rightarrow IC_k \rightarrow {}^p H^0(j_{k*} \mathbb{L}_{X_n^k}[\dim X_n^k]) \rightarrow {}^p H^0(j_{k+1*} \mathbb{L}_{X_n^{k+1}}[\dim X_n^{k+1}]) \rightarrow 0$$

and

$$0 \rightarrow {}^p H^0(j_{k+1*} \mathbb{L}_{X_n^{k+1}}[\dim X_n^{k+1}]) \rightarrow {}^p H^0(j_{k*} \mathbb{L}_{X_n^k}[\dim X_n^k]) \rightarrow IC_k \rightarrow 0$$

Proof: Let $g_k : X_n^k \hookrightarrow U_k = X_n^k \cup X_n^{k+1}$ be an open embedding and $h_k : X_n^{k+1} \hookrightarrow U_k$ be the complementary closed embedding. We get a distinguished triangle

$$h_{k*} h_k^!(IC_k|_{U_k}) \rightarrow IC_k|_{U_k} \rightarrow g_{k*} \mathbb{L}_{X_n^k}[\dim X_n^k] \rightarrow .$$

Let $f_k : U_k \hookrightarrow \overline{X_n^k}$ be the inclusion map. Applying the pushforward f_{k*} to the above distinguished triangle we get

$$j_{k+1*} h_k^! (\mathrm{IC}_k|_{U_k}) \rightarrow f_{k*} \mathrm{IC}_k|_{U_k} \rightarrow j_{k*} \mathbb{K}_{X_n^k}[\dim X_n^k] \rightarrow .$$

Taking perverse cohomology, we get the following long exact sequence

$$\begin{aligned} {}^p H^0(j_{k+1*} h_k^! (\mathrm{IC}_k|_{U_k})) &\rightarrow {}^p H^0(f_{k*} \mathrm{IC}_k|_{U_k}) \rightarrow {}^p H^0(j_{k*} \mathbb{K}_{X_n^k}[\dim X_n^k]) \\ &\rightarrow {}^p H^1(j_{k+1*} h_k^! (\mathrm{IC}_k|_{U_k})) \rightarrow {}^p H^1(f_{k*} \mathrm{IC}_k|_{U_k}) \rightarrow \end{aligned}$$

Using the Lemma 3.8 and Theorem 3.9, we get the desired result. Applying Verdier duality, we get the other short exact sequence.

Corollary 3.11: For $0 \leq k \leq n$, the perverse sheaf ${}^p H^0(j_{k*} \mathbb{K}_{X_n^k}[\dim X_n^k])$ (respectively ${}^p H^0(j_k! \mathbb{K}_{X_n^k}[\dim X_n^k])$) is uniserial with composition factors $\mathrm{IC}_k, \dots, \mathrm{IC}_n$ (respectively $\mathrm{IC}_n, \dots, \mathrm{IC}_k$).

Lemma 3.12: The perverse sheaf ${}^p H^0(j_0! \mathbb{K}_{X_n^0}[\dim X_n^0])$ (respectively ${}^p H^0(j_0* \mathbb{K}_{X_n^0}[\dim X_n^0])$) is the projective cover (respectively the injective envelope) of IC_0 .

Proof. Since j_0 is an open embedding, the functor j_0^* is t -exact and we have

$$\begin{aligned} \mathrm{Hom}({}^p H^0(j_0! \mathbb{K}_{X_n^0}[\dim X_n^0]), {}^p H^0(j_0! \mathbb{K}_{X_n^0}[\dim X_n^0])) &= \mathrm{Hom}({}^p \tau^{\geq 0}(j_0! \mathbb{K}_{X_n^0}[\dim X_n^0]), {}^p H^0(j_0! \mathbb{K}_{X_n^0}[\dim X_n^0])) \\ &= \mathrm{Hom}(j_0! \mathbb{K}_{X_n^0}[\dim X_n^0], {}^p H^0(j_0! \mathbb{K}_{X_n^0}[\dim X_n^0])) \\ &= \mathrm{Hom}(\mathbb{K}_{X_n^0}[\dim X_n^0], j_0^*({}^p H^0(j_0! \mathbb{K}_{X_n^0}[\dim X_n^0]))) \\ &= \mathrm{Hom}(\mathbb{K}_{X_n^0}[\dim X_n^0], {}^p H^0(j_0^* j_0! \mathbb{K}_{X_n^0}[\dim X_n^0])) \\ &= \mathrm{Hom}(\mathbb{K}_{X_n^0}[\dim X_n^0], \mathbb{K}_{X_n^0}[\dim X_n^0]) \cong \mathbb{k}. \end{aligned}$$

Thus, the object ${}^p H^0(j_0! \mathbb{K}_{X_n^0}[\dim X_n^0])$ is indecomposable.

Moreover,

$$\begin{aligned} \mathrm{Hom}((j_0! \mathbb{K}_{X_n^0}[\dim X_n^0]), \mathrm{IC}_k[1]) &= \mathrm{Hom}(\mathbb{K}_{X_n^0}[\dim X_n^0], j_0^! \mathrm{IC}_k[1]) \\ &= \mathrm{Hom}(\mathbb{K}_{X_n^0}[\dim X_n^0], j_0^* \mathrm{IC}_k[1]) \end{aligned}$$

and

$$j_0^* \mathrm{IC}_k[1] = \begin{cases} \mathbb{K}_{X_n^0}[\dim X_n^0 + 1], & \text{if } k = 0 \\ 0, & \text{otherwise} \end{cases}.$$

From the Lemma 3.5, we get $\mathrm{Hom}(j_0! \mathbb{K}_{X_n^0}[\dim X_n^0], \mathrm{IC}_k[1]) = 0$ for all $0 \leq k \leq n$.

Consider the distinguished triangle

$$j_0! \mathbb{K}_{X_n^0}[\dim X_n^0] \rightarrow {}^p H^0(j_0! \mathbb{K}_{X_n^0}[\dim X_n^0]) \rightarrow {}^p \tau^{<0} j_0! \mathbb{K}_{X_n^0}[\dim X_n^0][1] \rightarrow$$

Applying the functor $\mathrm{Hom}(-, \mathrm{IC}_k[1])$ to the distinguished triangle, we get the following long exact sequence

$$\begin{aligned} \mathrm{Hom}({}^p \tau^{<0} j_0! \mathbb{K}_{X_n^0}[\dim X_n^0], \mathrm{IC}_k) &\rightarrow \mathrm{Ext}^1({}^p H^0(j_0! \mathbb{K}_{X_n^0}[\dim X_n^0]), \mathrm{IC}_k) \\ &\rightarrow \mathrm{Hom}(j_0! \mathbb{K}_{X_n^0}[\dim X_n^0], \mathrm{IC}_k[1]) \rightarrow . \end{aligned}$$

Since ${}^p\tau^{<0}j_{0!}\mathbb{K}_{X_n^0}[\dim X_n^0] \in {}^pD^{<0}$ and $IC_k \in {}^pD^{\geq 0}$, we get

$$\mathrm{Hom}({}^p\tau^{<0}j_{0!}\mathbb{K}_{X_n^0}[\dim X_n^0], IC_k) = 0.$$

Hence, $\mathrm{Hom}(j_{0!}\mathbb{K}_{X_n^0}[\dim X_n^0], IC_k[1]) = 0$ implies

$$\mathrm{Ext}^1({}^pH^0(j_{0!}\mathbb{K}_{X_n^0}[\dim X_n^0]), IC_k) = 0.$$

Using the Theorem 3.10, we see ${}^pH^0(j_{0!}\mathbb{K}_{X_n^0}[\dim X_n^0])$ has a unique simple quotient i.e., IC_0 . Hence, it is the projective cover of IC_0 . Using Verdier duality, $\mathbb{D}({}^pH^0(j_{0!}\mathbb{K}_{X_n^0}[\dim X_n^0])) \cong {}^pH^0(j_{0*}\mathbb{K}_{X_n^0}[\dim X_n^0])$ we get the injective envelope of IC_0 . $\square$

We can also construct the injective envelope and the projective cover of IC_n .

Corollary 3.13: Let $\mathbb{T}$ be the Fourier–Sato transform functor as in . Then the perverse sheaf $\mathbb{T}({}^pH^0(j_{0!}\mathbb{K}_{X_n^0}[\dim X_n^0]))$ (respectively $\mathbb{T}({}^pH^0(j_{0*}\mathbb{K}_{X_n^0}[\dim X_n^0]))$) is the projective cover (respectively the injective envelope) of IC_n .

Proof. The functor $\mathbb{T}$ is an equivalence of categories that preserves projectives, injectives and simples. It also takes the constant sheaf IC_0 to the skyscraper sheaf IC_n . Since ${}^pH^0(j_{0!}\mathbb{K}_{X_n^0}[\dim X_n^0])$ is the projective cover and ${}^pH^0(j_{0*}\mathbb{K}_{X_n^0}[\dim X_n^0])$ is the injective envelope of IC_0 , we get our result. $\square$

Remark 3.14: Using the Corollary 3.11 and the Corollary 3.13, the multiplicity of IC_k in the projective cover of IC_0 and the projective cover of IC_n is one.

Using the Lemma 3.12 and the Corollary 3.11, the composition series of the projective cover of IC_0 gives us the following path in the associated quiver:

$$0 \xrightarrow{u_0} 1 \xrightarrow{u_1} 2 \xrightarrow{u_2} \dots \xrightarrow{u_{n-1}} n.$$

Note that there is only one arrow between consecutive vertices because the multiplicity of each IC_k in the projective cover of IC_0 i.e., ${}^pH^0(j_{0!}\mathbb{K}_{X_n^0}[\dim X_n^0])$ is one. By Verdier duality, there must also be an arrow in the opposite direction, i.e.,

$$0 \xleftarrow[v_0]{u_0} 1 \xleftarrow[v_1]{u_1} 2 \xleftarrow[v_2]{u_2} \dots \xleftarrow[v_{n-1}]{u_{n-1}} n$$

From the Theorem 3.6 and the Theorem 3.7, there are no loops and no arrows between non consecutive vertices. Thus, at the level of underlying quivers, we have found all the arrows.

4 Quiver with relations

Let Q be a quiver and $\mathbb{k}Q$ be the path algebra with basis given by directed paths in Q and multiplication induced by concatenation of paths.

Definition 4.1: A *relation* in Q is a $\mathbb{k}$ -linear combination of paths of length at least two having the same source and target. A *quiver with relations* $\widehat{Q} := (Q, I)$ is a quiver Q with a finite set of relations I .

The quiver algebra of $\widehat{Q}$ is the quotient $\mathbb{k}Q/\langle I \rangle$ of the path algebra by the ideal generated by the relations. We will assume that the ideal of relations $\langle I \rangle$ contains all path whose length is large enough, so that the corresponding quiver algebra is finite dimensional.

Example 4.2: The quiver $\widehat{AA}_n$ is the quiver with $n+1$ vertices, arrows and relations as follows,

$$0 \begin{array}{c} \xrightarrow{u_0} \\ \xleftarrow{v_0} \end{array} 1 \begin{array}{c} \xrightarrow{u_1} \\ \xleftarrow{v_1} \end{array} 2 \begin{array}{c} \xrightarrow{u_2} \\ \xleftarrow{v_2} \end{array} \cdots \begin{array}{c} \xrightarrow{u_{n-1}} \\ \xleftarrow{v_{n-1}} \end{array} n$$

with the relations

$$u_i v_i = 0 = v_i u_i.$$

Lőrincz and Walther [LW19] showed the category of G_n equivariant D -modules on X_n is equivalent to the category of representations of $\widehat{AA}_n$.

We have the quiver as described in the above section but we should also describe relations between arrows u_i 's and v_i 's. On the level of quiver, it matches with $\widehat{AA}_n$. We will show that it also has the same relations.

5 The main theorem

Let $\text{Perv}_{G_n}(X_n, \mathbb{k})_n$ be the full subcategory of $\text{Perv}_{G_n}(X_n, \mathbb{k})$ consisting of G_n -equivariant perverse sheaves on X_n that do not have IC_n as a composition factor. We will try to understand this subcategory and how much information it contains to understand $\text{Perv}_{G_n}(X_n, \mathbb{k})$.

Let $X_{1,1}$ be the principal open subset of matrices in X_n whose $(1,1)$ entry is nonzero and

$$H := \left\{ \begin{pmatrix} x & 0 \\ v & A \end{pmatrix} \times \begin{pmatrix} y & w \\ 0 & B \end{pmatrix} \mid x, y \in \mathbb{C}^\times, v, w^T \in \mathbb{C}^{n-1} \text{ and } (A, B) \in G_{n-1} \right\}$$

be a subgroup of G_n . Observe that $X_n^n = \{0\}$ is the only orbit contained entirely in the complement of $X_{1,1}$. Moreover, the subset $X_{1,1}$ is H -stable which gives the following:

Lemma 5.1: Let $j : X_{1,1} \hookrightarrow X_n$ be the inclusion map. Then

$$j^* : \text{Perv}_{G_n}(X_n, \mathbb{k})_n \hookrightarrow \text{Perv}_H(X_{1,1}, \mathbb{k}) \quad (5)$$

is fully faithful.

Proof. Exactness holds because j is an open embedding. Let $F, G \in \text{Perv}_{G_n}(X_n, \mathbb{k})_n$. Let i be the complementary closed embedding to j and consider the distinguished triangle,

$$i_! i^! G \rightarrow G \rightarrow j_* j^* G \rightarrow .$$

Applying the functor $\text{Hom}(F, -)$, we get

$$\text{Hom}(F, i_! i^! G) \rightarrow \text{Hom}(F, G) \rightarrow \text{Hom}(F, j_* j^* G) \rightarrow \text{Hom}(F, i_! i^! G[1]) \rightarrow .$$

Observe that ${}^p H^0(i^! G), {}^p H^0(i^* F)$ are supported on the complement of $X_{1,1}$. The only orbit contained entirely in the complement of $X_{1,1}$ is X_n^n which implies that the only simple subobject it may contain is IC_n . Since $F, G \in \text{Perv}_{G_n}(X_n, \mathbb{k})_n$, we get

$$\begin{aligned} {}^p H^0(i^! G) &= 0 \\ {}^p H^0(i^* F) &= 0. \end{aligned}$$

This shows $i^! G \in {}^p D^{\geq 1}$ and $i^* F \in {}^p D^{\leq -1}$.

By adjunction, and since i is closed embedding,

$$\text{Hom}(F, i_! i^! G[k]) = \text{Hom}(i^* F, i^! G[k]) = 0$$

for $k = 0, 1$. Hence,

$$\text{Hom}(F, G) \cong \text{Hom}(F, j_* j^* G)$$

Using adjunction for j_* i.e.,

$$\text{Hom}(F, G) \cong \text{Hom}(F, j_* j^* G) \cong \text{Hom}(j^* F, j^* G),$$

we prove the result. $\square$

Theorem 5.2: There is a fully faithful functor

$$F : \text{Perv}_{G_n}(X_n, \mathbb{k})_n \hookrightarrow \text{Perv}_{G_{n-1}}(X_{n-1}, \mathbb{k}).$$

Proof. We embed X_{n-1} in $X_{1,1}$ by sending $M \in X_{n-1}$ to $\begin{pmatrix} 1 & 0 \\ 0 & M \end{pmatrix} \in X_{1,1}$.

Set $G'_n := G_{n-1} \times \mathbb{C}^\times$ and embed it in H by sending $(A, B, c) \in G'_n$ to $\begin{pmatrix} c & 0 \\ 0 & A \end{pmatrix} \times \begin{pmatrix} c & 0 \\ 0 & B \end{pmatrix} \in H$. Since $\mathbb{C}^\times \subset H$ acts trivially on the copy of $X_{n-1} \subset X_{1,1}$, we get

$$\text{Perv}_{G_{n-1}}(X_{n-1}, \mathbb{k}) \cong \text{Perv}_{G'_n}(X_{n-1}, \mathbb{k}). \quad (6)$$

One can check that the natural map

$$\begin{aligned} \phi : H \times_{G'_n} X_{n-1} &\rightarrow X_{1,1} \\ (h, X) &\mapsto h \cdot X \end{aligned}$$

is well defined and an isomorphism as H -variety, i.e.,

$$H \times_{G'_n} X_{n-1} \cong X_{1,1} \quad (7)$$

. Using the inductive equivalence [Ach21, Theorem 6.5.10], j^* is fully faithful and the above isomorphism, we obtain an embedding of the category F as shown below

$$\begin{array}{ccccc} \text{Perv}_{G_n}(X_n, \mathbb{k})_n & \xrightarrow{5} & \text{Perv}_H(X_{1,1}, \mathbb{k}) & \xrightarrow{7} & \text{Perv}_H(H \times_{G'_n} X_{n-1}, \mathbb{k}) \\ & & & & \downarrow \text{Induction equivalence} \\ & & & & \text{Perv}_{G'_n}(X_{n-1}, \mathbb{k}) \\ & \searrow \text{embedding of category } F & & & \downarrow 6 \\ & & & & \text{Perv}_{G_{n-1}}(X_{n-1}, \mathbb{k}) \end{array}$$

□

Theorem 5.3: Let $X_n = M_n(\mathbb{C})$ and $G_n = GL_n(\mathbb{C}) \times GL_n(\mathbb{C})$. Then there is an equivalence of categories

$$\text{Perv}_{G_n}(X_n, \mathbb{k}) \cong \text{rep}(\widehat{AA_n})$$

for all $n \geq 1$.

Proof. We will prove this statement by induction on n . It is easy to check for $n = 1$. Let $n \geq 2$ and assume $\text{Perv}_{G_{n-1}}(X_{n-1}, \mathbb{k}) \cong \text{rep}(\widehat{AA_{n-1}})$. The quiver associated to $\text{Perv}_{G_n}(X_n, \mathbb{k})_n$ is obtained by removing the n^{th} vertex from the quiver associated to $\text{Perv}_{G_n}(X_n, \mathbb{k})$ but keeping the same relations i.e.,

$$0 \begin{array}{c} \xrightarrow{u_0} \\ \xleftarrow{v_0} \end{array} 1 \begin{array}{c} \xrightarrow{u_1} \\ \xleftarrow{v_1} \end{array} 2 \begin{array}{c} \xrightarrow{u_2} \\ \xleftarrow{v_2} \end{array} \cdots \begin{array}{c} \xrightarrow{u_{n-2}} \\ \xleftarrow{v_{n-2}} \end{array} n-1.$$

The functor F constructed in the Theorem 5.2 is fully faithful and the categories $\text{Perv}_{G_n}(X_n, \mathbb{k})_n$ and $\text{Perv}_{G_{n-1}}(X_{n-1}, \mathbb{k})$ have the same quiver. Thus, we get

$$\text{Perv}_{G_n}(X_n, \mathbb{k})_n \cong \text{Perv}_{G_{n-1}}(X_{n-1}, \mathbb{k}).$$

We get the relations

$$u_i v_i = 0 = v_i u_i$$

for $0 \leq i \leq n-2$ and no other relations amongst them.

Since $\mathbb{T}$ is an equivalence of categories sending IC_0 to IC_n , the relations $u_0 v_0 = 0 = v_0 u_0$ force the relation

$$u_{n-1} v_{n-1} = 0 = v_{n-1} u_{n-1}.$$

Thus, the quiver of $\text{Perv}_{G_n}(X_n, \mathbb{k})$ has all the relations as the quiver $\widehat{AA_n}$

It remains to rule out further relations involving u_{n-1} and v_{n-1} . The only possible ones are of the form $u_{n-1} u_{n-2} \cdots u_k = 0$ and $v_k v_{k+1} \cdots v_{n-1} = 0$. In either cases multiplicity of IC_k in the projective cover of IC_n decreases to 0 which contradicts the Remark 3.14. Thus,

$$u_i v_i = 0 = v_i u_i$$

for $0 \leq i \leq n-1$ are all the relations in our quiver associated to $\text{Perv}_{G_n}(X_n, \mathbb{k})$. □

6 Applications

The functor $\mathbb{T}$ takes the constant sheaf IC_0 to the skyscraper sheaf IC_n , but in general it is hard to compute $\mathbb{T}(\text{IC}_k)$ for $0 < k < n$.

Let S^k be the simple representation in $\text{rep}(\widehat{AA_n})$ associated to IC_k in $\text{Perv}_{G_n}(X_n, \mathbb{k})$. Using our main result, we can try describe indecomposable projective and injective objects from the quiver side. The projective cover of S^n is uniserial with composition factors $S^0, S^1, \dots, S^n$. This implies that the projective cover of IC_n i.e., $\mathbb{T}({}^p H^0(j_{0!} \mathbb{k}_{X_n^0}[\dim X_n^0]))$ is uniserial with composition factors $\text{IC}_0, \text{IC}_1, \dots, \text{IC}_n$.

From 3.11, the perverse sheaf $\mathbb{T}({}^p H^0(j_{0!}\mathbb{K}_{X_n^0}[\dim X_n^0]))$ is uniserial with composition factors $\mathbb{T}(\mathrm{IC}_n), \dots, \mathbb{T}(\mathrm{IC}_0)$. The projective cover being uniserial and matching the composition factors, we get

$$\mathbb{T}(\mathrm{IC}_k) \cong \mathrm{IC}_{n-k}$$

Remark 6.1: For $0 \leq k \leq n$, using the Corollary 3.11 the perverse sheaf $\mathbb{T}({}^p H^0(j_{k*}\mathbb{K}[\dim X_n^k]))$ is uniserial with composition factors $\mathrm{IC}_{n-k}, \dots, \mathrm{IC}_0$ and $\mathbb{T}({}^p H^0(j_{k!}\mathbb{K}[\dim X_n^k]))$ is uniserial with composition factors $\mathrm{IC}_0, \dots, \mathrm{IC}_{n-k}$.

6.1 Projective cover

We can construct the projective cover of IC_k understanding the properties of the projective cover from quiver side.

Remark 6.2: In $\widehat{AA_n}$, the multiplicity of each simple S^l in the projective cover of S^k is one. The projective cover of S^k is the following representation

$$\mathbb{K} \xleftarrow{1} \mathbb{K} \xleftarrow{1} \dots \xleftarrow{1} \mathbb{K} \xrightarrow{1} \dots \xrightarrow{1} \mathbb{K} \xrightarrow{1} \mathbb{K}$$

where the source is at vertex k .

For $0 \leq k \leq n$, let

$$j_0^k : \bigsqcup_{i=0}^k X_n^i \hookrightarrow X_n$$

be the open embedding of the union of first $k+1$ strata and

$$j_k : X_n^k \hookrightarrow X_n$$

be the inclusion map of the stratum X_n^k in X_n .

Lemma 6.3: For $t < k$,

$$\begin{aligned} {}^p H^0(j_{0!}j_0^{k*} \mathrm{IC}_t) &\cong \mathrm{IC}_t \\ {}^p H^0(j_{0*}j_0^{k*} \mathrm{IC}_t) &\cong \mathrm{IC}_t \\ j_{0!}^k(j_0^{k*} \mathrm{IC}_t) &\cong \mathrm{IC}_t. \end{aligned}$$

Proof. Let $i_0^k : \bigsqcup_{i=k+1}^n X_n^i \hookrightarrow X_n$ be the closed embedding. Using the following distinguished triangle,

$$j_{0!}^k j_0^{k*} \mathrm{IC}_t \longrightarrow \mathrm{IC}_t \longrightarrow i_{0*}^k i_0^{k*} \mathrm{IC}_t,$$

we get the following long exact sequence

$${}^p H^{-1}(i_{0*}^k i_0^{k*} \mathrm{IC}_t) \longrightarrow {}^p H^0(j_{0!}^k j_0^{k*} \mathrm{IC}_t) \rightarrow \mathrm{IC}_t \rightarrow {}^p H^0(i_{0*}^k i_0^{k*} \mathrm{IC}_t) \rightarrow .$$

Using 3 and t -exactness of i_{0*}^k ,

$${}^p H^l(i_{0*}^k i_0^{k*} \mathrm{IC}_t) = 0$$

for $l > -4$ which proves the first isomorphism. Applying Verdier duality to the first isomorphism, we get the second isomorphism. Using 3, 4, and [Ach21, Lemma 3.3.34] we get the third isomorphism. $\square$

Lemma 6.4: Let K be a perverse sheaf such that K does not have IC_t for $t \geq k$ as a composition factor. Then,

$${}^p H^0(j_{0!}^k j_0^{k*} K) \cong j_{0!*}^k (j_0^{k*} K).$$

Proof. We will prove this using induction on number of composition factors in K . When K is simple i.e., $K \cong \mathrm{IC}_t$. Since $t < k$, we use the Lemma 6.3.

Suppose K has n composition factors. Then there exists an exact sequence

$$0 \rightarrow \mathrm{IC}_t \rightarrow K \rightarrow K' \rightarrow 0$$

such that K' has $n - 1$ composition factors. Since K does not have IC_t for $t \geq k$ as a composition factor, then so is for K' . Thus, ${}^p H^0(j_{0!}^k j_0^{k*} K') \cong j_{0!*}^k (j_0^{k*} K')$. Using [Ach21, Lemma 3.3.34], we prove our lemma. $\square$

For $0 \leq k \leq n$, consider the perverse sheaves

$$\mathcal{P}_k := {}^p H^0(j_{0!}^k j_0^{k*} (\mathbb{T}({}^p H^0(j_{n-k!} \mathbb{K}_{X_n^{n-k}} [\dim X_n^{n-k}])))$$

The next lemma gives the multiplicity of IC_t 's in $\mathcal{P}_k$'s.

Lemma 6.5: For $0 \leq k \leq n$ and $0 \leq t \leq n$, the multiplicity of IC_t in $\mathcal{P}_k$ is one.

Proof. Let K be the kernel of the map $\mathbb{T}({}^p H^0(j_{n-k!} \mathbb{K}_{X_n^{n-k}} [\dim X_n^{n-k}]]) \rightarrow \mathbb{T}(\mathrm{IC}_{n-k}) \cong \mathrm{IC}_k$. Applying the right exact functor ${}^p H^0(j_{0!}^k j_0^{k*} (-))$ to the exact sequence

$$0 \longrightarrow K \longrightarrow \mathbb{T}({}^p H^0(j_{n-k!} \mathbb{K}_{X_n^{n-k}} [\dim X_n^{n-k}]]) \longrightarrow \mathrm{IC}_k \longrightarrow 0,$$

we get a long exact sequence

$$\longrightarrow {}^p H^0(j_{0!}^k j_0^{k*} K) \longrightarrow \mathcal{P}_k \longrightarrow {}^p H^0(j_{0!}^k j_0^{k*} \mathrm{IC}_k) \longrightarrow 0.$$

From the Corollary 3.11 and the Remark 6.1, the kernel K does not contain IC_t for $t \geq k$, using the Lemma 6.4 we also get exactness on left i.e.,

$$0 \longrightarrow j_{0!*}^k j_0^{k*} K \longrightarrow \mathcal{P}_k \longrightarrow {}^p H^0(j_{0!}^k j_0^{k*} \mathrm{IC}_k) \longrightarrow 0$$

is an exact sequence.

Using the Remark 6.1, the kernel K has composition factors IC_t for $t < k$ with multiplicity one. Using the Lemma 6.3, we get that the multiplicity of IC_t in $j_{0!*}^k j_0^{k*} K$ for $t < k$ is one.

Using the Corollary 3.11, the object

$${}^p H^0(j_{0!}^k j_0^{k*} \mathrm{IC}_k) \cong {}^p H^0(j_{k!} \mathbb{K}_{X_n^k} [\dim X_n^k])$$

has composition factors IC_t for $t \geq k$ with multiplicity one. Thus the multiplicity of IC_t in for $0 \leq t \leq n$ in $\mathcal{P}_k$ is one. $\square$

Lemma 6.6: For $0 \leq k \leq n$, the perverse sheaf $\mathcal{P}_k$ has a unique simple quotient IC_k .

Proof. Observe that for $0 \leq t \leq n$, we have

$$\begin{aligned} \text{Hom}(\mathcal{P}_k, IC_t) &\cong \text{Hom}({}^p H^0(j_{0!}^k j_0^{k*} (\mathbb{T}({}^p H^0(j_{n-k!} \mathbb{L}_{X_n^{n-k}} [\dim X_n^{n-k}])))), IC_t) \\ &\cong \text{Hom}({}^p \tau^{\geq 0}(j_{0!}^k j_0^{k*} (\mathbb{T}({}^p H^0(j_{n-k!} \mathbb{L}_{X_n^{n-k}} [\dim X_n^{n-k}])))), IC_t) \\ &\cong \text{Hom}(j_{0!}^k j_0^{k*} (\mathbb{T}({}^p H^0(j_{n-k!} \mathbb{L}_{X_n^{n-k}} [\dim X_n^{n-k}])))), IC_t) \\ &\cong \text{Hom}(\mathbb{T}({}^p H^0(j_{n-k!} \mathbb{L}_{X_n^{n-k}} [\dim X_n^{n-k}])), j_{0*}^k j_0^{k*} IC_t) \\ &\cong \text{Hom}(\mathbb{T}({}^p H^0(j_{n-k!} \mathbb{L}_{X_n^{n-k}} [\dim X_n^{n-k}])), {}^p H^0(j_{0*}^k j_0^{k*} IC_t)). \end{aligned}$$

For $t > k$, we have $j_{0!}^k j_0^{k*} IC_t = 0$ which implies

$$\text{Hom}(\mathcal{P}_k, IC_t) = 0.$$

For $t < k$, we have ${}^p H^0(j_{0*}^k j_0^{k*} IC_t) \cong IC_t$. Thus,

$$\text{Hom}(\mathcal{P}_k, IC_t) \cong \text{Hom}(\mathbb{T}({}^p H^0(j_{n-k!} \mathbb{L}_{X_n^{n-k}} [\dim X_n^{n-k}])), IC_t) = 0$$

since $\mathbb{T}({}^p H^0(j_{n-k!} \mathbb{L}_{X_n^{n-k}} [\dim X_n^{n-k}]))$ has a unique simple quotient IC_k .

For $t = k$, we have

$$\text{Hom}(\mathcal{P}_k, IC_k) \cong \text{Hom}(\mathbb{T}({}^p H^0(j_{n-k!} \mathbb{L}_{X_n^{n-k}} [\dim X_n^{n-k}])), {}^p H^0(j_{k*} \mathbb{L}_{X_n^k} [\dim X_n^k])).$$

Using the Corollary 3.11 and the Remark 6.1, $\mathbb{T}({}^p H^0(j_{n-k!} \mathbb{L}_{X_n^{n-k}} [\dim X_n^{n-k}]))$ has a unique simple quotient IC_k and IC_k is the unique simple subobject for ${}^p H^0(j_{k*} \mathbb{L}_{X_n^k} [\dim X_n^k])$. Since IC_k appears in both with multiplicity one, we get $\text{Hom}(\mathcal{P}_k, IC_k) = \mathbb{k}$. □

The following theorem proves $\mathcal{P}_k$ is the projective cover of IC_k .

Theorem 6.7: For $0 \leq k \leq n$, the perverse sheaf $\mathcal{P}_k$ is the projective cover of IC_k in $\text{Perv}_{G_n}(X_n, \mathbb{k})$.

Proof. Let P_k be the projective cover of IC_k in $\text{Perv}_{G_n}(X_n, \mathbb{k})$. From the Lemma 6.6, the perverse sheaf $\mathcal{P}_k$ has a unique simple quotient IC_k which gives a surjective map from P_k to $\mathcal{P}_k$.

$$\begin{array}{ccc} P_k & \dashrightarrow & \mathcal{P}_k \\ & \searrow & \downarrow \\ & & IC_k \end{array}$$

Using the Remark 6.2 and the Lemma 6.5, the multiplicity of simples matches in both perverse sheaves. The surjective map between perverse sheaves with same composition factors is an isomorphism which proves our theorem. □

References

- [Ach21] Pramod N Achar. *Perverse sheaves and applications to representation theory*. Vol. 258. American Mathematical Soc., 2021.
- [ASS06] Ibrahim Assem, Andrzej Skowronski, and Daniel Simson. *Elements of the representation theory of associative algebras: Volume 1: Techniques of representation theory*. Vol. 65. Cambridge University Press, 2006.
- [Bei+18] Alexander Beilinson et al. *Faisceaux pervers*. Vol. 4. Société mathématique de France Paris, 2018.
- [BG99] Tom Braden and Mikhail Grinberg. “Perverse sheaves on rank stratifications”. In: *Duke Math. J.* 96.2 (1999), pp. 317–362. ISSN: 0012-7094,1547-7398. DOI: 10.1215/S0012-7094-99-09609-6. URL: <https://doi-org.libezp.lib.lsu.edu/10.1215/S0012-7094-99-09609-6>.
- [CW22] Alessio Cipriani and Jon Woolf. “When are there enough projective perverse sheaves?” In: *Glasg. Math. J.* 64.1 (2022), pp. 185–196. ISSN: 0017-0895,1469-509X. DOI: 10.1017/S0017089521000021. URL: <https://doi-org.libezp.lib.lsu.edu/10.1017/S0017089521000021>.
- [GMV96] Sergei Gelfand, Robert MacPherson, and Kari Vilonen. “Perverse sheaves and quivers”. In: *Duke Math. J.* 83.3 (1996), pp. 621–643. ISSN: 0012-7094,1547-7398. DOI: 10.1215/S0012-7094-96-08319-2. URL: <https://doi-org.libezp.lib.lsu.edu/10.1215/S0012-7094-96-08319-2>.
- [LW19] András C. Lőrincz and Uli Walther. “On categories of equivariant $\mathcal{D}$ -modules”. In: *Adv. Math.* 351 (2019), pp. 429–478. ISSN: 0001-8708,1090-2082. DOI: 10.1016/j.aim.2019.04.051. URL: <https://doi-org.libezp.lib.lsu.edu/10.1016/j.aim.2019.04.051>.
- [MS74] John Willard Milnor and James D Stasheff. *Characteristic classes*. 76. Princeton university press, 1974.
- [Vil94] K. Vilonen. “Perverse sheaves and finite-dimensional algebras”. In: *Trans. Amer. Math. Soc.* 341.2 (1994), pp. 665–676. ISSN: 0002-9947,1088-6850. DOI: 10.2307/2154577. URL: <https://doi-org.libezp.lib.lsu.edu/10.2307/2154577>.